MATH FOUNDATIONS

Information Theory Visual Guide

Entropy, cross-entropy, and KL divergence —
how ML measures surprise and compresses reality.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Surprisal: Rare Events Carry More Bits

Information content of an event = -log2 P(event).

P = 1/2

coin flip -> 1 bit of surprise

P = 1/1024

rare event -> 10 bits of surprise

'The sun rose today' tells you ~0 bits. 'It snowed in July' tells you a lot.

Bits measure answers to yes/no questions. Surprise is the currency of information.

STEP 2/6

Entropy: Expected Surprise

H = -sum p(x) log2 p(x) — the average bits per symbol.

FAIR COIN

H = 1 bit — maximum uncertainty

ALWAYS HEADS

H = 0 bits — no uncertainty at all

MAX ENTROPY = uniform

a loaded die has LOWER entropy than a fair one

Entropy is the fundamental limit of compression: you cannot beat H bits per symbol on average.

STEP 3/6

Cross-Entropy: Coding With the Wrong Model

H(p, q): average bits when the world is p but you assumed q.

TRUE p

reality's distribution

MODEL q

your approximation

H(p,q) >= H(p)

mismatch always costs extra bits — equality only if q = p

Training a classifier = shrinking the gap between your q and the world's p.

STEP 4/6

KL Divergence: The Price of Being Wrong

D_KL(p || q) = H(p,q) - H(p) — extra bits from using q instead of p.

D_KL(p || q) >= 0, = 0 iff p = q

a directed distance between distributions

NOT SYMMETRIC

D(p||q) != D(q||p) — direction matters

MODE-SEEKING

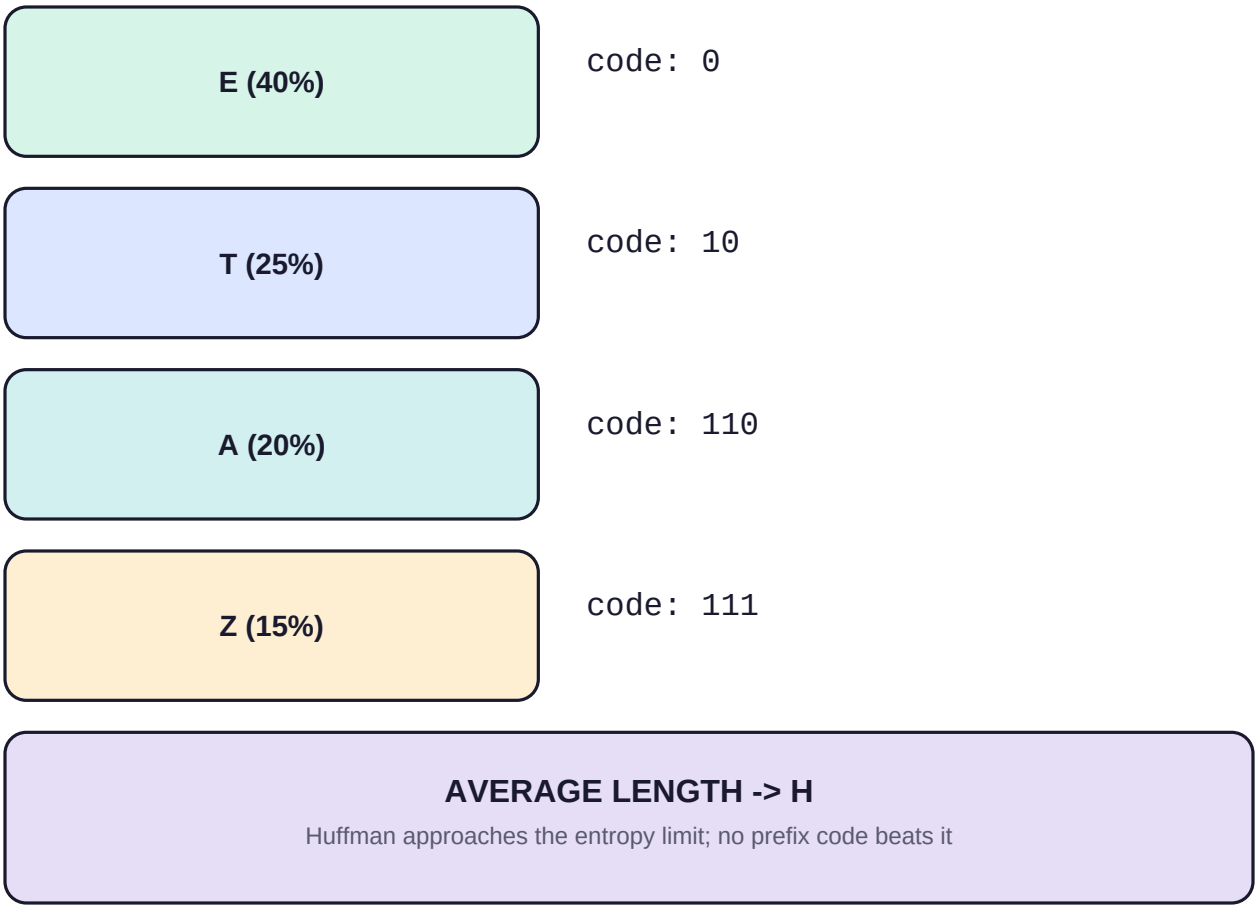
variational inference minimizes D_KL to fit q to p

Not a true distance (asymmetric), but the workhorse divergence of ML.

STEP 5/6

Huffman Coding: Entropy Made Practical

Frequent symbols get short codes; rare ones get long codes.



STEP 6/6

Where ML Uses All of This

Your loss function is probably information theory.

Log loss

classification loss IS cross-entropy between labels and predictions

Decision trees

splits chosen to maximize information gain (entropy drop)

VAEs / diffusion

KL terms keep learned distributions near a prior

Mutual information

I(X;Y): how much X tells you about Y — feature selection

STEP ✓I✓

Key Takeaways

Information Theory — Remember

- Surprisal (-log p) measures how informative an event is; rare events carry more bits.
- Entropy is expected surprise — the theoretical floor for compression.
- Cross-entropy measures cost under your model q when reality is p; H(p,q) >= H(p).
- KL divergence is the extra-bit price of model mismatch; zero only when q = p.
- Huffman coding assigns short codes to frequent symbols, approaching the entropy limit.
- Log loss, information gain, and VAE objectives are information theory wearing an ML costume.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev